

```

1  #include <stdio.h>
2
3  int main()
4  {
5      int A[100][4];
6      int i, j, n, total = 0, index, temp;
7      float avg_wt, avg_tat;
8
9      printf("Enter number of process: ");
10     scanf("%d", &n);
11
12     printf("Enter Burst Time:\n");
13     for (i = 0; i < n; i++)
14     {
15         printf("P%d: ", i + 1);
16         scanf("%d", &A[i][1]);
17         A[i][0] = i + 1;
18     }
19
20     // Sorting processes by Burst Time (SJF)
21     for (i = 0; i < n; i++)
22     {
23         index = i;
24         for (j = i + 1; j < n; j++)
25         {
26             if (A[j][1] < A[index][1])
27                 index = j;
28         }
29
30         // Swap burst time
31         temp = A[i][1];
32         A[i][1] = A[index][1];
33         A[index][1] = temp;
34
35         // Swap process id
36         temp = A[i][0];
37         A[i][0] = A[index][0];
38         A[index][0] = temp;
39     }
40
41     // Waiting Time calculation
42     A[0][2] = 0;
43     for (i = 1; i < n; i++)
44     {
45         A[i][2] = 0;
46         for (j = 0; j < i; j++)
47         {
48             A[i][2] += A[j][1];
49         }
50         total += A[i][2];
51     }

```

```

P1: 12
P2: 14
P3: 15
P4: 16

```

| P  | BT | WT | TAT |
|----|----|----|-----|
| P1 | 12 | 0  | 12  |
| P2 | 14 | 12 | 26  |
| P3 | 15 | 26 | 41  |
| P4 | 16 | 41 | 57  |

```

Average Waiting Time = 19.750000
Average Turnaround Time = 34.000000

```

```

...Program finished with exit code 0
Press ENTER to exit console.

```